

challenges/scripts/ host_no_auto_suspend_challenge.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:** Active
CONST-033 host-hardening reproduction guard. Self-contained (no framework dependency). Verifies the DEVELOPER HOST cannot auto-suspend.

Companion (§11.4.18) to the in-source documentation block at the top of the script.

Overview / Purpose

CONST-033 forbids any host-level power-state transition (auto-suspend has historically caused data loss on the mission-critical host). This challenge asserts that the host it runs on cannot be suspended / hibernated / put into hybrid-sleep by any user, session, desktop environment, greeter, or cron job — "defence in depth". It checks the runtime host state produced by the sibling `install-host-suspend-guard.sh` fix (target masking + `sleep.conf` override + `logind IdleAction` override), not the source tree (that is `no_suspend_calls_challenge.sh`).

Four assertions (all must pass):

1. `systemctl is-enabled` reports masked for **all four** of `sleep.target`, `suspend.target`, `hibernate.target`, `hybrid-sleep.target`.
2. `AllowSuspend=no` is present in `/etc/systemd/sleep.conf` or any `/etc/systemd/sleep.conf.d/*.conf` drop-in.
3. `logind IdleAction` is `ignore` (or `unset`, which defaults to `ignore`).
4. The journal shows zero `The system will suspend now` broadcasts since the fix marker `/etc/systemd/sleep.conf.d/00-no-suspend.conf` was written.

Usage

`bash challenges/scripts/host_no_auto_suspend_challenge.sh`

No arguments. Run it after `install-host-suspend-guard.sh` has hardened the host. It reads system state only — it never changes power settings.

Inputs

- None on the command line.
- Reads host state: `systemctl` is-enabled for the four sleep targets; `/etc/systemd/sleep.conf` + `/etc/systemd/sleep.conf.d/*.conf`; `/etc/systemd/logind.conf` + `/etc/systemd/logind.conf.d/*.conf`; the fix marker `/etc/systemd/sleep.conf.d/00-no-suspend.conf`; and `journalctl` since the marker's mtime.

Outputs

- Human-readable `PASS: / FAIL:` lines per assertion, a per-target state listing, and a `=== summary: N pass, M fail ===` line on stdout.
- Exit code: 0 = all 4 assertions `PASS`; 1 = one or more `FAIL`; 2 = invocation error (documented in-source; not currently reached by any code path).

Side-effects

None. Purely read-only host introspection — no files written, no services touched, no power state changed. (§11.4.14 quiescent — the target host is left exactly as found.)

Dependencies

`bash`, `systemctl`, `journalctl`, `grep`, `stat`, `date`, `tr`, `cut`, `head` (`systemd` host). Absent optional inputs degrade to `<unset>` / unknown rather than crashing (`set -uo pipefail` with guarded command substitutions, §11.4.1).

Edge cases

- **Fix marker missing** → assertion 4 FAILs with a pointer to run `install-host-suspend-guard.sh`.
- **A sleep target not masked** → assertion 1 FAILs, listing the offending target(state).
- **IdleAction unset** is treated as safe (`systemd` defaults to ignore).

- Assertion 4 dates the window from the marker's mtime, so pre-fix suspend broadcasts do not count against the current hardening.

Related scripts

- `scripts/host-power-management/install-host-suspend-guard.sh` — the fix this challenge validates.
- `challenges/scripts/no_suspend_calls_challenge.sh` — the SOURCE-tree gate (this one is the HOST-state gate).
- `scripts/host-power-management/check-no-suspend-calls.sh` — the static scanner.
- Constitution CONST-033 (host power-management hard ban), §11.4.14.

Last verified

2026-07-01 — documented against the script source; `sh -n / bash -n parse-clean`. The live host-state assertions run on a `systemd` host per CONST-033 verification.